DMPNN Performance Estimates Diverge across Evaluation Boundaries in a Cyclic-Peptide Permeability Benchmark

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Abstract

Machine-learning models for cyclic-peptide permeability are often evaluated using splits that place records from the same source or analogue series on both sides of the test boundary. Before version-2 fitting, we froze an internal protocol that grouped 6,895 curated parallel artificial membrane permeability assay (PAMPA) records into source/analogue-connected blocks and confined model development to outer-training data. The same directed message-passing neural network (DMPNN) achieved source-macro mean absolute errors of 0.467 under molecule-grouped pseudorandom evaluation and 1.005 under joint blocking (difference 0.537; 95% confidence interval, 0.324–0.753); CR1 and delete-one-block intervals remained positive, and the fixed weighting-plus-descriptor candidate did not close the gap. Because joint blocking also changed training-set profiles, the comparison quantifies divergence between the two procedures rather than an isolated dependence-removal effect; for this dataset and DMPNN, molecule-grouped performance therefore did not establish transfer to jointly held-out sources and analogue families.

Keywords: cyclic peptides; membrane permeability; data leakage; cross-validation; domain shift

Introduction

Cyclic peptides combine extended binding surfaces with chemical tunability, making them attractive for targets that are difficult to address with conventional small molecules. Their size, polarity, and conformational complexity, however, create a narrow and sequence-dependent window for passive membrane permeation. Experimental and simulation studies have linked permeability to conformational flexibility, intramolecular hydrogen bonding, lipophilicity, polar-group exposure, and environment-dependent conformational ensembles [1, 2, 3, 4]. These factors are not captured by a single local motif and can change across solvents, topologies, and residue modifications. Predictive modeling is therefore appealing both for prioritizing compounds and for testing which molecular representations retain information about permeability.

Public permeability compilations have made larger retrospective studies possible. CycPeptMPDB consolidated measurements reported across publications and patents and provided a common resource for model development [5]. Graph neural networks, multilevel molecular features, peptide language models, multimodal architectures, and solvent-dependent conformational ensembles have subsequently achieved strong performance on retrospective partitions of cyclic-peptide data [6, 7, 8, 9]. These advances broaden the available representation toolkit. At the same time, a compiled database is not a single homogeneous experiment. Its records inherit differences in assay implementation, reporting practice, chemical campaign, and analogue density from their original sources. A large row count can therefore overstate the number of independent prediction settings represented by the data.

The evaluation split determines which of these settings a performance estimate describes. A molecule-random split estimates interpolation when training and test records are sampled from the same compiled mixture. It does not necessarily estimate transfer to a new experimental source or a new analogue family. Records from one publication can share assay offsets and reporting conventions, whereas compounds from one medicinal-chemistry series can share scaffold-specific structure–property relationships. If either dependency crosses the training-test boundary, the test set may contain close contextual or chemical neighbors of the training data. This is different from direct outcome leakage: the labels may remain formally hidden, but the resampling unit is still smaller than the unit over which independence is scientifically required.

Molecular benchmarks have addressed related problems with scaffold, chronological, and simulated time splits [10, 11, 12, 13]. Dependence-aware cross-validation, analogue-bias analyses, and similarity-aware splitting likewise show that the held-out unit should follow the dependence structure and intended deployment setting [14, 15, 16]. Reporting guidance for machine learning in the life sciences consequently emphasizes provenance, optimization boundaries, model selection, and evaluation design rather than treating a split label as a sufficient description of validity [17]. For cyclic peptides, a conventional Murcko scaffold is also an incomplete dependence unit because topology, stereochemistry, residue-level edits, and noncanonical monomers can define close series that are not expressed by a single two-dimensional scaffold label.

Existing cyclic-peptide permeability studies have considered molecule-random, scaffold, representation-cluster, source-held-out, canonical-group, and simulated out-of-distribution partitions [18, 19, 7, 8]. These protocols provide important and increasingly demanding tests (Supplementary Table S6), but source provenance and analogue connectivity answer different questions. Holding out a source while allowing its analogue series to remain through another source preserves a chemical dependence path. Holding out an analogue component while retaining records from the same experimental source preserves a provenance path. The two grouping variables must therefore be connected before splitting. Their connected components, rather than either variable alone, define the smallest blocks that simultaneously exclude complete sources and transitive analogue families.

This joint construction also exposes the effective evidence geometry. Thousands of molecular records can collapse into only a few independent connected blocks, and those blocks can differ by orders of magnitude in size. Evaluation must then preserve the blocks while preventing the largest source-rich component from defining every summary. Nested selection is equally important: a grouped outer test is not sufficient if preprocessing, early stopping, hyperparameter selection, or interval calibration has already used records connected to that test block. A leakage-resistant protocol must apply the same boundary throughout model development.

Here we quantify how much the estimated accuracy of one reproduced DMPNN on one public cyclic-peptide PAMPA benchmark differs across two evaluation boundaries. We froze a peptide-aware analogue graph and a source-provenance graph, formed 18 connected joint blocks, and evaluated every block once in an outer leave-one-joint-block-out design. All preprocessing, model selection, early stopping, and residual calibration were confined to the other 17 blocks. The primary internally prespecified hypothesis compared joint-block and molecule-grouped pseudorandom source-macro error. A second confirmatory hypothesis tested whether a fixed source/component-balanced and descriptor-augmented DMPNN improved prediction and empirical interval coverage under joint shift. By retaining negative outcomes and preventing post hoc candidate replacement, the study evaluates a bounded generalization claim rather than proposing a universally superior split or a new state-of-the-art predictor.

Methods

Study design and endpoint

This retrospective study evaluated passive permeability measured by the parallel artificial membrane permeability assay (PAMPA). The continuous endpoint was log10-transformed apparent permeability (Papp), with Papp expressed in cm s^{-1} before transformation. The source database contained 7,298 PAMPA rows. A row was classified as irrecoverably censored when either source detection-limit field was populated; under the rule frozen before version-2 modeling, all 372 such rows were excluded from the primary continuous endpoint. This left 6,926 uncensored rows with parseable structures and sequences. Records were then grouped by data source and isomeric canonical SMILES. A multi-row group was eligible for collapse only when year, database version, topology signature, and canonical sequence agreed across its uncensored records; its endpoint was the median of the eligible measurements. Thirty-one compatible multi-row groups were collapsed, yielding 6,895 analytical source–structure records representing 6,862 unique stereochemical molecules from 41 data sources. No incompatible group or invalid structure/sequence was present in this frozen population. A release-safe row manifest preserves raw identifiers, censoring flags, inclusion or linked-exclusion status, and curated identifiers without redistributing structures or permeability labels (Supplementary Table S1).

An outcome-independent audit showed that the 372 censored rows arose in 23 raw-data sources. Of these rows, 259 carried database-assigned or reported floor values, 40 explicit upper limits, 19 no reportable value, 9 non-detection or below-limit notes, 7 solubility or non-testing failures, and 38 other detection-limit notes. Only two censored rows were linked to a retained group that also contained an uncensored replicate; the remaining 370 had no retained continuous-endpoint record. The raw table did not contain structured fields for PAMPA pH, membrane composition, incubation time, temperature, or donor/acceptor conditions. Source identity therefore represents a combined provenance and protocol proxy rather than a known categorical assay-condition variable (Supplementary Table S9).

Curation retained stereochemistry, cyclic topology, source provenance, and the frozen endpoint definition. Molecular standardization and the analogue relation were completed without using permeability values. No exclusion depended on a prediction, residual, model rank, or held-out-block performance. An earlier version-1 partition was permanently retired after its feasibility assumptions failed; neither that partition nor its label vault had an analytical role in the present study. All reported results use the version-2 population, graph, fold manifests, thresholds, models, and decision rules frozen before the first version-2 fit.

The study contained two sequential confirmatory hypotheses. H1 asked whether the D0 baseline had higher source-macro mean absolute error (MAE) under joint source/analogue-block evaluation than under molecule-grouped pseudorandom five-fold cross-validation. Support required a joint-minus-random difference of at least 0.10 log10(Papp) and a two-sided 95% interval entirely above zero. H2 evaluated D3, the only fixed model-improvement candidate, under joint shift. Support required all five internally prespecified conditions: improvements of at least 0.03 versus D0 and 0.05 versus the nested-selected classical procedure; a paired block-bootstrap interval excluding zero in the favorable direction; pooled nominal 90% interval coverage between 85% and 95%; and source-macro Spearman correlation no more than 0.03 below the best comparator. Failure of any condition rejected H2, and secondary ablations could not replace D3 after outcome release.

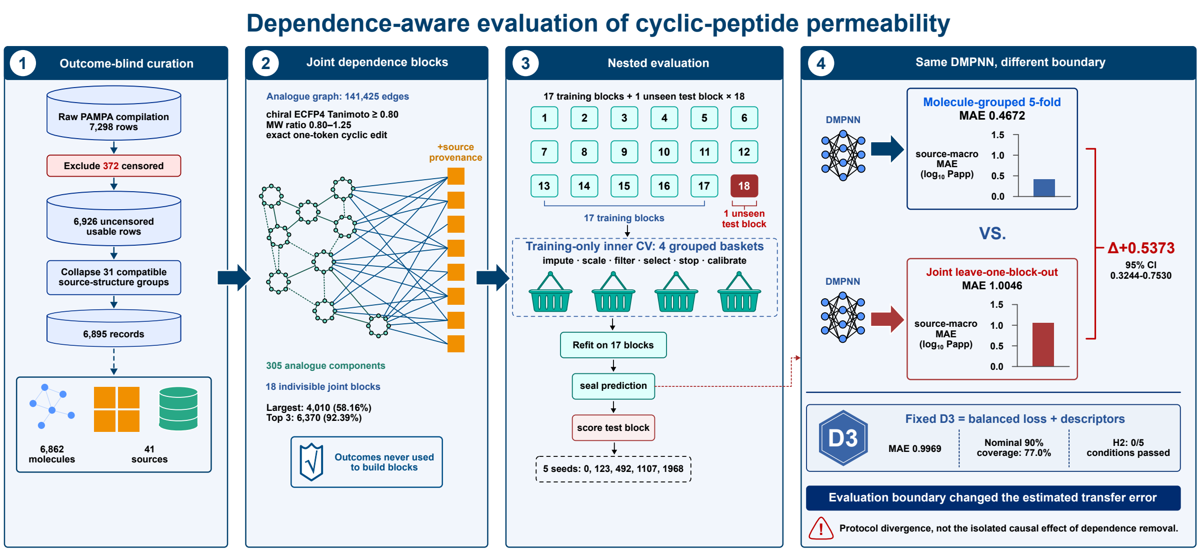

Figure 1: Dependence-aware evaluation of cyclic-peptide permeability. 1, Outcome-blind curation of the cyclic-peptide PAMPA compilation. Of 7,298 raw rows, 372 carrying detection-limit annotations were excluded, leaving 6,926 uncensored usable rows. Median collapse of 31 compatible multi-row source-structure groups produced 6,895 analytical records representing 6,862 unique molecules from 41 sources. **2,** Peptide-aware analogue relationships were closed transitively into 305 analogue components and combined with source provenance in a bipartite graph. Connected components produced 18 indivisible joint blocks; neither permeability values nor model errors were used to construct them. The largest block contained 4,010 records (58.16%) and the three largest contained 6,370 records (92.39%). **3,** Each joint block served once as an unseen outer test block while the other 17 blocks formed the outer-training population. Imputation, scaling, filtering, selection, stopping and residual calibration were restricted to four training-only grouped inner baskets before refitting, prediction sealing and test-block scoring. **4,** Under the completed molecule-grouped five-fold and joint leave-one-block-out procedures, the same D0 DMPNN had source-macro MAEs of 0.4672 and 1.0046, respectively; the joint-minus-molecule-grouped difference was 0.5373 (95% percentile interval, 0.3244-0.7530). The panel also summarizes the fixed D3 candidate (MAE 0.9969; nominal-90% slot-pooled interval coverage 77.0%; 0/5 H2 conditions passed). The comparison describes divergence between the two completed procedures and does not isolate the causal contribution of dependence removal from the changed training-size profile. Icons and block widths are conceptual and are not drawn to scale.

Protocol status and deviations

The version-2 research map, cross-validation governance, experiment plan, and hypothesis document were frozen internally on 17 July 2026 before any version-2 fit. They were not deposited in an independently time-stamped public registry before outcome release. We therefore describe the hypotheses and decision rules as prospectively frozen or internally prespecified rather than as independently public preregistration.

The original plan specified a D0 comparison ladder comprising molecule-grouped pseudorandom, chirality-aware Murcko, source-only, analogue-component-only, and joint-block evaluations. The confirmatory joint-block and molecule-grouped arms were completed, but the Murcko, source-only, and analogue-only diagnostic arms were not completed before H1/H2 outcome release. On 10 August 2026, an outcome-blind operational plan froze source-only and analogue-only manifests together with a newly added size-profile-matched molecule-random control. Those future runs are post-confirmatory and cannot alter the original H1/H2 decisions. The Murcko arm remains an uncompleted protocol item and is not represented as a reported analysis (Supplementary Table S10).

Table 1 summarizes the resulting data and evaluation geometry. The distinction between molecular rows and independent joint blocks is central to interpretation: the primary metric gives each source equal weight, whereas uncertainty resampling preserves every record and source within a sampled joint block.

Table 1: Frozen data and evaluation geometry. Counts were determined before confirmatory model fitting. Joint blocks are connected components of the source–analogue-component graph and were never divided across training and test sets.

Quantity	Frozen value	Analytical role
Raw PAMPA rows	7,298	Starting database records
Censored rows excluded	372	Frozen continuous-endpoint rule
Uncensored usable raw rows	6,926	Input to source–structure grouping
Compatible multi-row groups collapsed	31	Median of uncensored same-source replicates
Curated source–structure groups	6,895	Confirmatory analytical records
Unique stereochemical molecules	6,862	Molecular population
Data sources	41	Units of source-macro aggregation
Frozen analogue edges	141,425	Peptide-aware similarity/edit graph
Analogue components	305	Transitive chemical-dependence units
Joint source/analogue blocks	18	Outer resampling and test units
Largest block	4,010 records (58.16%)	Preserved without subdivision
Three largest blocks	6,370 records (92.39%)	Dominant evidence geometry

Source and analogue graph

The primary analogue relation was frozen before confirmatory modeling. Molecular nodes were the 6,862 unique curated stereochemical structures. Two structures were connected when their chiral radius-2 ECFP4 Tanimoto similarity was at least 0.80 and their molecular-weight ratio was between 0.80 and 1.25. The molecular-weight constraint prevented a high fingerprint similarity alone from linking structures with markedly different overall size. A prespecified exact one-token cyclic-edit relation at unchanged topology and ring length supplemented the fingerprint rule so that close residue substitutions remained connected even when a global fingerprint threshold was not sufficient. The union of these relations contained 141,425 edges.

Transitive closure of the molecular graph produced 305 analogue components. Transitivity was retained because evaluation of a local series should not depend on whether two terminal compounds share a direct edge when both are connected through intervening analogues. We next constructed a bipartite graph with analogue components on one side and data sources on the other. An edge joined an analogue component to every source contributing one or more records to that component. Connected components of this bipartite graph were defined as joint blocks. This operation closes alternating dependence paths such as source A–analogue series 1–source B–analogue series 2, which would remain open under either source-only or analogue-only grouping.

The procedure yielded 18 indivisible joint blocks containing 4,010; 1,518; 842; 249; 105; 36; 18; 18; 17; 16; 16; 11; 10; 8; 7; 7; 4; and 3 records (Figure 2a). The largest block contained 58.16% of the analytical population, and the three largest together contained 92.39% (Figure 2b). Their connectivity also differed sharply: the largest block linked 162 analogue components across 20 sources, whereas the 1,518-record and 842-record blocks each originated from one source but contained 3 and 112 analogue components, respectively (Figure 2c). These blocks were not rebalanced by splitting them, because doing so would reintroduce the very source or analogue paths the protocol was designed to exclude. Permeability values were unavailable to the graph and fold-construction procedures, and stable identifiers and serialized manifests were fixed before model fitting.

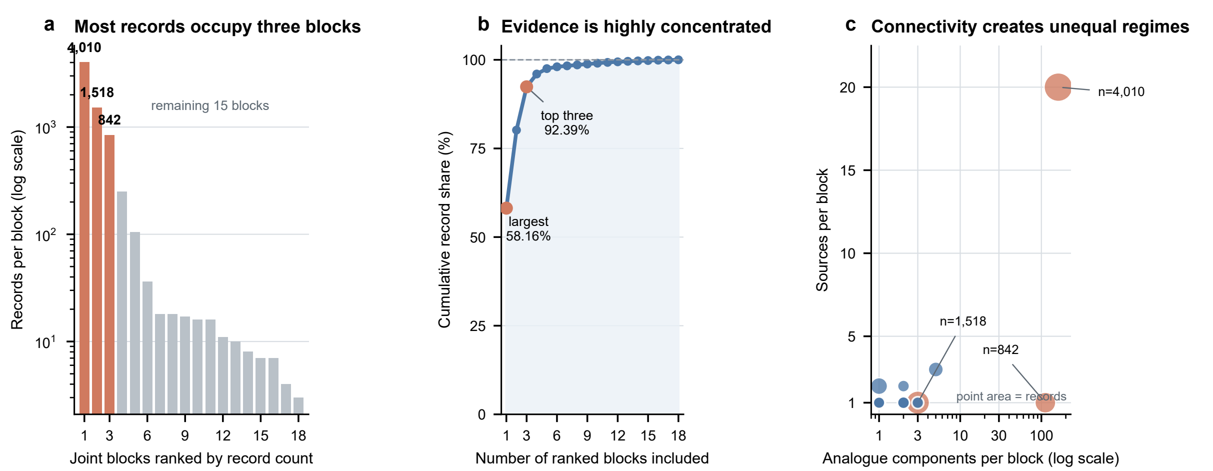

Figure 2: Effective evidence geometry of the frozen joint blocks. a, Record counts for all 18 joint blocks, ranked from largest to smallest and shown on a logarithmic axis. Orange denotes the three largest blocks. **b,** Cumulative fraction of the 6,895 analytical records captured by the ranked blocks. The largest block contains 58.16% of records and the three largest contain 92.39%. **c,** Number of analogue components and contributing sources in each block. Point area represents the number of curated records, and the three largest blocks are labelled. All points represent frozen blocks; no jitter or simulated values were used. These panels describe the dependence structure of the dataset rather than a balanced target population. Source data are provided in the figure source-data directory.

Nested joint-block evaluation

Each joint block served once as the outer test set, while the other 17 blocks formed the outer-training population (Figure 1). The procedure was deterministic after the graph freeze: blocks were sorted by stable identifiers, and every record received exactly one outer out-of-fold (OOF) prediction per model and seed. Feature imputation, scaling, feature filtering, hyperparameter selection, early stopping, and residual calibration were restricted to the corresponding outer-training population. Thus, neither the labels nor the covariate distribution of an outer block could determine a fitted transform or model-selection decision.

Within each outer fold, the 17 available blocks were assigned to four inner baskets without outcomes. Blocks were ordered by decreasing record count, with stable block identifiers breaking ties, and each block was placed into the currently smallest basket. This greedy rule reduced avoidable imbalance while keeping every joint block intact. The assignments were serialized and hashed before training. Classical hyperparameters were selected by mean inner source-macro MAE, using row-micro MAE and then lower compute as pre-specified tie-breakers. For DMPNN models, the architecture and optimizer configuration remained fixed; the inner folds determined only the stopping epoch. The rounded-up median of the four inner best epochs was used when refitting on all 17 outer-training blocks.

The same baskets also defined cross-fitted residual pools. For each fixed point model, four inner fits omitted one basket in turn and predicted that basket. Their absolute residuals were pooled within the outer fold, after which the point model was refitted on all outer-training blocks before predicting the outer test block. This construction prevented outer-test errors from contributing to interval width. Because only 18 complete joint blocks existed, the four-basket design was chosen as a computationally feasible compromise that preserved the full dependence boundary; exact leave-one-inner-block-out selection would have required 17 inner fits for every outer fold and configuration.

For H1, D0 was additionally evaluated by molecule-grouped pseudorandom five-fold cross-validation on the same 6,895 records and the same five seed indices. The indivisible assignment unit was the unique stereochemical `molecule_id`, so repeated source-specific records of the same molecule could not cross folds. Molecule groups were ordered by decreasing record count, with their stable structure-derived identifiers breaking ties, and were assigned to the currently smallest fold; all five folds contained 1,379 records. Sources and analogue components were permitted to cross this comparison boundary. The primary statistic was joint-block source-macro MAE minus molecule-random source-macro MAE. Source-macro MAE first averaged absolute error within each of the 41 data sources and then assigned equal weight to every source. Row-micro MAE and block-level summaries were secondary because pooled record weighting would allow the largest connected block and its largest sources to dominate the inference target.

Models and comparators

D0 was the locked reproduction of the BenchmarkCycPeptMP directed message-passing neural network at maintained commit `d82aa3c5c9c849dbd584e8669132ed3d33e50a27` [19]. The frozen implementation retained the published molecular graph featurizer, DMPNN architecture, Delaney pretraining, batch size of 64, optimizer and scheduler families, maximum of 2,000 epochs, and patience of 200. Five seed indices (0–4) mapped to effective random-number-generator seeds 0, 123, 492, 1107, and 1968. Model architecture, optimizer family, and learning-rate schedule were not searched on the joint-block benchmark. Only the stopping epoch was transferred from the grouped inner folds as described above.

The comparator ladder deliberately ranged from a no-structure predictor to nonlinear molecular models. The outer-training mean predicted every held-out record with the mean endpoint of the corresponding outer-training population. Ridge regression, Random Forest, and XGBoost used a 2,048-bit radius-2 ECFP concatenated with the frozen medicinal-chemistry descriptor panel. Ridge alpha was selected from 0.01, 0.1, 1, 10, and 100. Random Forest used 500 trees and a closed grid over maximum-feature fraction, minimum leaf size, and maximum depth. XGBoost used a closed grid over estimator count, depth, learning rate, and L2 regularization, with subsampling and feature subsampling fixed at 0.8. Hyperparameters were selected independently within each outer fold. The reported nested-selected classical comparator was therefore a prespecified fold-wise procedure, not whichever algorithm performed best after outer errors were viewed.

D1 tested whether the loss function, rather than the molecular representation, was responsible for poor transfer. It added source/component-balanced weighting to D0. Each source received equal total weight; within a source, each analogue component received equal weight; and records divided the component weight. This hierarchy prevented a record-rich source or dense local series from dominating the optimization objective. The weighting did not alter the evaluation metric or the outer split.

D2 tested a fixed chemistry/topology augmentation. A descriptor vector was concatenated at the DMPNN readout and encoded topology class, ring size, molecular weight, cLogP, topological polar surface area, hydrogen-bond donors and acceptors, rotatable bonds, formal charge, fraction sp3, stereocenter count, N-methyl count, noncanonical-residue count, and prespecified missingness flags. Descriptor definitions and missingness handling were frozen before confirmatory fitting. Continuous transformations were estimated only from outer-training records and then applied unchanged to the outer block.

D3 combined the D1 weighting and D2 descriptors and was the sole fixed H2 candidate. The design tested whether equalizing provenance/series contributions and supplying explicit medicinal-chemistry features produced complementary gains. D1 and D2 were mandatory secondary ablations that localized the contributions of the two switches. They could not replace D3 after the confirmatory outcomes were released, even if one displayed a lower point estimate.

Statistical analysis

All headline DMPNN results were summarized across five seeds. The primary point estimate for each model was the arithmetic mean of its five seed-specific source-macro MAEs. Absolute errors from tiny blocks and sources remained in MAE aggregation, but correlations were suppressed for a source with fewer than 10 eligible observations. This rule avoided reporting unstable rank statistics while retaining every record in the primary error analysis.

For H1, we generated 10,000 deterministic bootstrap replicates using the 18 outer blocks as the resampling unit. A replicate sampled 18 blocks with replacement, retained all records and sources inside each sampled block, and resampled the corresponding molecule-random result within seed. We recalculated joint-block and molecule-random source-macro MAEs and stored their difference. The two-sided 95% percentile interval was computed from this bootstrap distribution. Resampling molecules independently would have contradicted the dependence structure used to define the target estimand.

For H2 condition 3, nested-selected classical and D3 OOF predictions were paired by record, outer block, and seed. Ten thousand block/seed bootstrap replicates estimated classical-minus-D3 source-macro MAE; positive values favored D3. Source-macro Spearman correlation was calculated within each source containing at least 10 observations, averaged equally across the 29 eligible sources, and then averaged across seeds. H2 condition 5 compared D3 with the best D0 or classical value under this definition.

Cross-fitted empirical intervals used absolute inner-validation residuals from the selected D3 checkpoint for each outer fold. For a residual pool of size n and nominal level q, the half-width was the k-th ordered residual, where k = min(n, ceil((n + 1)q)). The same fold-local half-width was applied symmetrically to every D3 outer prediction from that fold. Coverage, mean full width, and observed-minus-nominal calibration error were reported at 50%, 80%, and 90%. The primary coverage denominator comprised 34,475 record-by-seed prediction slots: every curated record appeared once for each of five algorithmic seeds. These repeated seed realizations are not independent experimental units; we therefore report per-seed ranges and equal-source and equal-block descriptive aggregations rather than a binomial confidence interval. Molecular property studies have used Bayesian, ensemble, and conformal approaches to quantify predictive uncertainty [20, 21, 22]. Our construction also uses held-out residual quantiles, but we describe it as a cross-fitted empirical interval rather than an exact distribution-free split-conformal guarantee because an unseen joint block need not be exchangeable with the inner residual pool.

Post-confirmatory robustness and sensitivity analyses

After the confirmatory H1 result was released, we performed descriptive analyses without fitting or selecting a model. First, each joint block was omitted in turn and the D0 source-macro gap was recalculated over the remaining sources. We also recalculated the gap under row-micro weighting and equal-block-macro weighting, where each block contributed the mean of its source-level MAEs. These analyses tested influence and estimand sensitivity but did not replace the internally prespecified block/seed bootstrap.

To address the small number and unequal sizes of the 18 clusters, we additionally averaged the five seeds within each source and fitted an intercept-only CR1 cluster-robust variance estimator with sealed block as cluster and a t critical value with 17 degrees of freedom. We calculated a delete-one-block jackknife standard error with the same t reference and exhaustively enumerated all 2^18 block-level sign flips of the paired source contribution. The sign-flip test assumes independent blocks and joint sign symmetry under the null; the CR1 and jackknife calculations assume the sealed blocks are the relevant independent clusters. These were post-confirmatory sensitivities, not replacements for the internally prespecified bootstrap.

Second, D0 absolute and signed errors were summarized for every source and block across the five seeds. Point-prediction calibration used one five-seed mean OOF prediction per record. Records were assigned to ten equal-frequency prediction bins separately within each evaluation arm, and mean observed permeability was compared with mean predicted permeability. Descriptive calibration slopes were obtained by least-squares regression of observed values on mean OOF predictions. No post-hoc recalibration was fitted.

Third, the outcome-blind analogue and source/component graphs were reconstructed at the prespecified descriptive Tanimoto thresholds of 0.70 and 0.90, alongside the primary 0.80 threshold. Chiral ECFP4 settings, the 0.80-1.25 molecular-weight ratio, exact one-token cyclic-edit rule, source provenance and all curated records were held fixed. This analysis compared partition geometry only. The frozen OOF predictions were not rescored under alternative partitions because their training boundaries did not correspond to those partitions.

Execution integrity and reproducibility

The frozen D123 execution plan contained 551 stages, including 491 scientific model fits, and produced accepted artifacts for every scheduled stage. Execution used one local NVIDIA GPU under a recorded software environment. Append-only, hash-chained ledgers linked each scientific stage to its input hashes, dependencies, configuration, attempt namespace, and output manifest. Interrupted or failed attempts remained preserved rather than being overwritten, and a rerun could be accepted only under the same frozen scientific identity. Confirmatory metric payloads were opened only after prediction sealing, hash verification, and an explicit release record.

The public reporting boundary was separated from the private label boundary. Prediction artifacts recorded whether labels were loaded only after prediction sealing, and the released D1–D3 payloads each contained the expected 34,475 OOF prediction slots (6,895 records × 5 seeds). Automated checks verified zero joint-block overlap in outer folds, zero source and analogue-component overlap under the primary protocol, one OOF slot per record/model/seed, and training-only fitting of transformations and stopping decisions.

Row-level inner-validation residuals had not been persisted during the original fits, so interval reporting used a separately governed checkpoint-only reconstruction. The analysis restored all 72 accepted D3 inner checkpoints (18 outer folds × 4 inner baskets) with fitting, optimizer updates, and checkpoint writes disabled. Model-state hashes were identical before and after every inference pass. The reconstruction produced 117,215 fold-local residual slots with exact outer-training coverage; 71 checkpoints came from their first accepted attempt and one from a documented second attempt. No model was retrained, tuned, replaced, or selected during this reporting analysis.

Results

Molecule-random evaluation substantially understated joint-block error

Under molecule-grouped pseudorandom five-fold evaluation, D0 achieved a source-macro MAE of 0.4672. Under joint source/analogue-block leave-one-block-out evaluation, the same baseline achieved 1.0046 (Figure 3a). The joint-block-minus-random difference was 0.5373 log10(Papp), more than five times the internally prespecified 0.10 support threshold. Relative to the molecule-grouped estimate, source-macro error was approximately 115% higher under the joint protocol. This percentage describes the two fitted evaluation procedures and does not isolate dependence removal from their different training-size profiles.

The 10,000-replicate block/seed bootstrap 95% confidence interval was 0.3244–0.7530 and lay entirely above zero. Both H1 conditions therefore passed. The lower confidence limit alone exceeded three times the minimum effect specified before model fitting, indicating that support did not depend on a narrowly positive interval. Because blocks rather than molecules were resampled, the interval also retained the dependence structure targeted by the evaluation. With only 18 highly unequal blocks, however, the interval should be interpreted as uncertainty over the observed benchmark geometry rather than a high-precision population interval.

This comparison changed the scientific interpretation of D0 without changing the analytical population, endpoint, architecture, five seeds, or model capacity. Molecule-random evaluation suggested a sub-half-log-unit error after equal weighting of sources. The joint-block protocol yielded an error of approximately one log unit, corresponding to a much weaker claim about transfer to an unseen source and analogue family. Because the joint folds also had a more variable training-size profile than the five random folds, the observed difference combines the dependence boundary with fold-specific training-set reduction; it does not by itself identify their separate contributions. A post-confirmatory, size-profile-matched molecule-random analysis was frozen before any new fit to address this limitation, but its results are not included in this version.

The joint-block result must also be read against the effective sample structure. Although the analysis contained 6,895 records, 58.16% belonged to one connected block and 92.39% to the three largest blocks. H1 consequently estimates performance over the observed collection of 18 connected regimes; it does not imply that 6,895 independent chemical series were evaluated.

Post-confirmatory analyses supported a broad but heterogeneous H1 contrast

No single joint block accounted for the qualitative H1 result (Supplementary Fig. S1a). Omitting each block in turn produced source-macro gaps from 0.4724 to 0.5523 log10(Papp), and all 18 remained above the internally prespecified 0.10 point-gap threshold. Simultaneously omitting the three largest blocks left only 525 records across 19 sources and 15 blocks, but the descriptive gap remained 0.5335. This extreme stress test changed the represented population and was not treated as a new hypothesis test.

Small-cluster analyses reached the same directional conclusion while making their assumptions explicit (Supplementary Table S3). The CR1 sealed-block-clustered interval was 0.3777–0.6970, and the delete-one-block jackknife interval was 0.3779–0.6968. Block-specific mean gaps were positive for 17 of 18 blocks. Exhaustive block-level sign flipping gave a two-sided p value of 1.53×10^{-5} (4 of 262,144 assignments as or more extreme). These post-confirmatory calculations show that the observed contrast was not generated by one high-leverage block, but they do not increase the number of independent regimes beyond 18.

The direction also persisted across weighting schemes (Supplementary Fig. S1c). The joint-block-minus-random gap was 0.3502 under row-micro aggregation, 0.5373 under the primary source-macro aggregation, and 0.4662 when each block received equal weight. These estimands differed in magnitude because records, sources and blocks were highly unequal, but none restored the random-split estimate under joint shift.

The contrast was distributed across most sources rather than confined to a small subset (Supplementary Fig. S2a). Mean joint-block MAE exceeded molecule-random MAE for 38 of 41 sources. The difference was positive in all five seeds for 28 sources and in at least four seeds for 36. The largest source-specific differences often occurred in small sources, so their magnitudes were treated as localization diagnostics rather than stable independent effects.

Joint shift also broadened source-specific signed bias and degraded point-prediction calibration (Supplementary Fig. S2b,c). Median absolute source bias increased from 0.0784 under molecule-random evaluation to 0.5779 under joint-block evaluation. Absolute bias exceeded 0.5 log10(Papp) for 3 of 41 random-split source summaries and 23 of 41 joint-block summaries. The observed-on-predicted calibration slope decreased from 0.9637 to 0.5950, while decile-weighted absolute calibration error increased from 0.0269 to 0.1808.

Alternative analogue thresholds changed molecular components more than joint-block concentration (Supplementary Fig. S3). Thresholds of 0.70, 0.80 and 0.90 produced 91, 305 and 745 analogue components, respectively, but only 15, 18 and 20 source/component joint blocks. The largest block retained 57.84-58.80% of all records, and the three largest retained 92.07-93.02%. Thus, evidence concentration was not specific to the primary threshold, although this geometry analysis did not test whether H1 performance was invariant to the alternative partitions.

The fixed D3 modification did not improve prediction under joint shift

The five-seed mean source-macro MAEs were 1.0046 for D0, 0.9128 for the nested-selected classical procedure, 0.9050 for D1, 0.9708 for D2, and 0.9969 for D3 (Figure 3b). D3 improved over D0 by only 0.0077, below the required 0.03. It was also 0.0841 worse than the classical procedure rather than at least 0.05 better. H2 conditions 1 and 2 therefore failed.

The ablations separated the two proposed modifications. Balanced loss alone reduced source-macro MAE by 0.0996 relative to D0, whereas descriptor augmentation alone reduced it by only 0.0337. Combining the changes did not add their gains: D3 returned close to D0 and was 0.0919 worse than D1. D1 was slightly better than the nested-selected classical procedure by 0.0079, but this difference was descriptive and below the prespecified H2 margins. D1 could not replace D3 because the candidate identity had been fixed before outcome review.

Metric aggregation exposed an additional limitation. Row-micro MAEs were 0.6430 for D0, 0.6477 for the nested-selected classical procedure, 0.6478 for D1, 0.7459 for D2, and 0.8597 for D3. Thus, D0 appeared competitive when every record had equal weight even though its source-macro MAE was the worst of the five displayed procedures. A descriptive outer-training mean predictor achieved a source-macro MAE of 0.7769 and a row-micro MAE of 0.6073, outperforming all learned models on these aggregate point metrics under joint shift. This result does not establish that constant prediction is generally preferable; it shows that added representation complexity did not overcome the source and analogue changes in this dataset.

The paired bootstrap contradicted a favorable D3 effect. The statistic was classical-minus-D3 source-macro MAE, for which positive values would favor D3. Its point estimate was -0.0841, and the 95% interval was -0.2316 to -0.0169. The interval excluded zero entirely in the direction of D3 degradation, so H2 condition 3 failed. Table 2 reports the complete internally prespecified decision rather than stopping after the first failed conditions.

Table 2: Complete internally prespecified H2 decision. H2 required all five conditions to pass. Improvements are defined so that positive values favor D3; the negative classical comparison therefore indicates degradation.

H2 condition	Observed result	Internally prespecified requirement	Decision
D3 improvement over D0	0.0077 MAE	$\geq$0.0300	Fail
D3 improvement over nested-selected classical	−0.0841 MAE	$\geq$0.0500	Fail
Paired block/seed bootstrap versus classical	95% CI −0.2316 to −0.0169	Favorable interval excluding 0	Fail
Nominal 90% empirical interval coverage	77.0%	85–95%	Fail
Spearman deficit from best comparator	0.0766	$\leq$0.0300	Fail

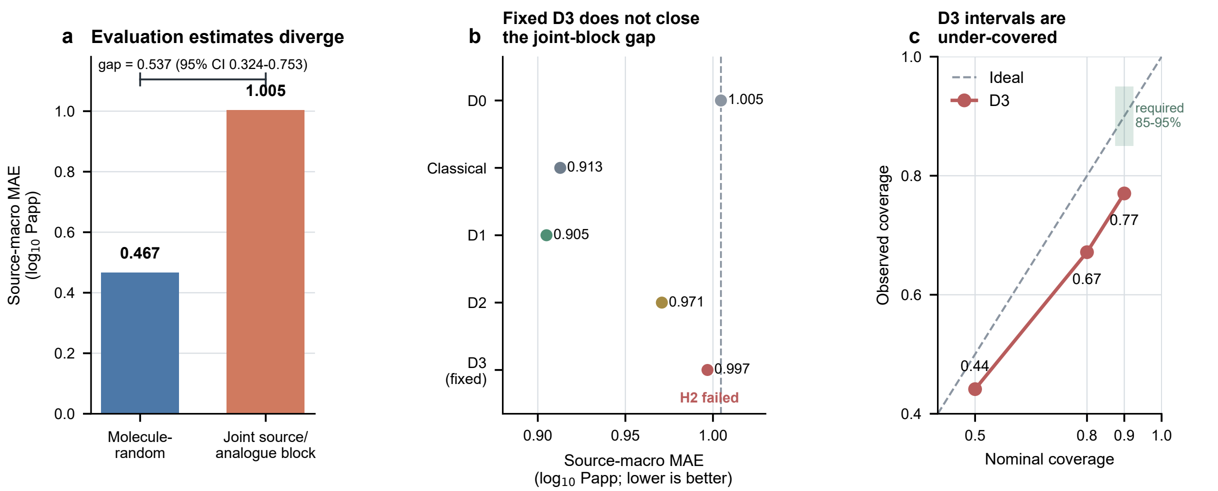

Figure 3: Performance estimates and calibration across evaluation procedures. a, Source-macro mean absolute error (MAE; log10 Papp) for the same D0 DMPNN under molecule-grouped pseudorandom and joint source/analogue-block evaluation. The annotated difference is joint-block minus molecule-grouped MAE; the 95% interval is the percentile interval from 10,000 paired outer-block/seed bootstrap replicates. Because the joint folds also changed the training-size profile, the contrast describes divergence between the two completed evaluation procedures rather than an isolated causal effect of dependence removal. **b,** Five-seed mean source-macro MAE under the joint-block protocol for D0, the nested-selected classical procedure, D1, D2 and the internally prespecified fixed candidate D3. Points show model means, and the dashed vertical line marks D0. Lower values indicate better prediction. D1 and D2 are secondary ablations and cannot replace D3 after outcome release. **c,** Nominal versus observed D3 empirical interval coverage across 34,475 frozen out-of-fold record-by-seed prediction slots (6,895 records x 5 seeds). Fold-local half-widths were computed from checkpoint-only inner-validation residuals using the finite-sample upper order statistic. The green segment marks the internally prespecified acceptable range (85-95%) for the nominal 90% interval. D3 achieves 77.0% slot-pooled coverage and is therefore under-covered; repeated seed slots are not independent experimental units. Source data are provided in the figure source-data directory.

D3 uncertainty intervals were under-covered

Observed D3 interval coverage was 44.1%, 67.2%, and 77.0% for nominal 50%, 80%, and 90% intervals, respectively (Figure 3c). The corresponding calibration errors were −5.9, −12.8, and −13.0 percentage points. Under-coverage therefore increased rather than diminished at the two higher nominal levels. The mean full widths were 1.023, 1.987, and 2.614 log10(Papp), showing that the 90% intervals were already broad in the endpoint scale yet still failed to include enough held-out outcomes.

Nominal 90% coverage fell below the internally prespecified acceptable range of 85–95%, so H2 condition 4 failed. Coverage was evaluated over all 34,475 frozen D3 OOF prediction slots. Across seeds, 90% coverage ranged from 75.8% to 78.1%; equal-source and equal-block descriptive coverage were 71.4% and 75.0%, respectively (Supplementary Table S11). Each interval used only the residual pool reconstructed for its own outer-training population, and no outer-block residual changed the interval applied to that block. The observed deficit therefore cannot be attributed to direct use of outer-test errors during calibration.

The reconstruction audit provided an additional integrity check on this negative result. All 72 D3 inner checkpoints were restored without fitting, and model-state hashes remained unchanged before and after inference. The 117,215 reconstructed residual slots exactly covered the required fold-local validation populations. Under-coverage consequently reflects the behavior of the frozen residual-quantile procedure under joint shift rather than a newly trained or retrospectively selected interval model.

D3 did not preserve source-level rank ordering

Across the 29 sources with at least 10 eligible observations, source-macro Spearman correlation was 0.1978 for D3, 0.2314 for D0, and 0.2744 for the nested-selected classical procedure. The classical procedure was therefore the best comparator for within-source rank ordering. D3 trailed it by 0.0766, more than twice the maximum permitted deficit of 0.03, so H2 condition 5 failed.

This criterion tested a different property from absolute error. A model can have a tolerable source-specific offset while still ordering analogues usefully within that source, or it can achieve a moderate MAE while failing to preserve ranks. D3 did not satisfy the relaxed requirement that its source-macro rank correlation remain close to the best comparator. Because H2 required all five conditions, and every condition failed, the fixed D3 hypothesis was not supported under any of its prespecified predictive or uncertainty criteria.

Failure structure was not explained by seed noise alone

D1 source-macro MAE across the five seeds was 0.8707, 0.8956, 0.9266, 0.8673, and 0.9645. D2 and D3 were worse than D1 at every corresponding seed (Figure 4a). The ordering was therefore not produced by one unfavorable random initialization, although the spread of D1 values confirmed that seed uncertainty remained material.

At source level, D1 outperformed D0 in 22 of 41 sources. D2 was worse than D1 in 19 sources, and D3 was worse than D1 in 22 (Figure 4b). The largest D3-versus-D1 degradations occurred for the sources labeled 2018_Lee (+1.3409 MAE), 2018_Buckton (+1.0551), 2021_Kelly (+0.6992), 2013_CHUGAI (+0.6709), and 2017_Pye (+0.5146). The first two were particularly informative because D1 had improved over D0 by 1.2602 and 0.9895 MAE, respectively, before descriptor augmentation largely erased those gains.

The same heterogeneity appeared across outer blocks. D1 beat D0 in 7 of 18 blocks. D3 was worse than D1 in 9 blocks and worse than D2 in 11 (Figure 4c). In the 1,518-record fold 3, MAE increased from 0.5641 for D1 to 1.2632 for D3; in the 842-record fold 7, it increased from 1.0533 to 1.7242. By contrast, the largest 4,010-record fold 5 did not collapse: D3 achieved 0.5292 compared with 0.5627 for D1.

These patterns are consistent with a regime-dependent negative interaction involving the descriptor pathway. They argue against a universal failure of balanced weighting and against a single-seed accident. They do not, however, identify whether descriptor distribution shift, feature scaling, optimization competition, or redundancy with the learned representation caused the interaction. Selecting descriptor subsets or a replacement model from these revealed OOF errors would turn the confirmatory benchmark into a tuning set and was therefore not performed.

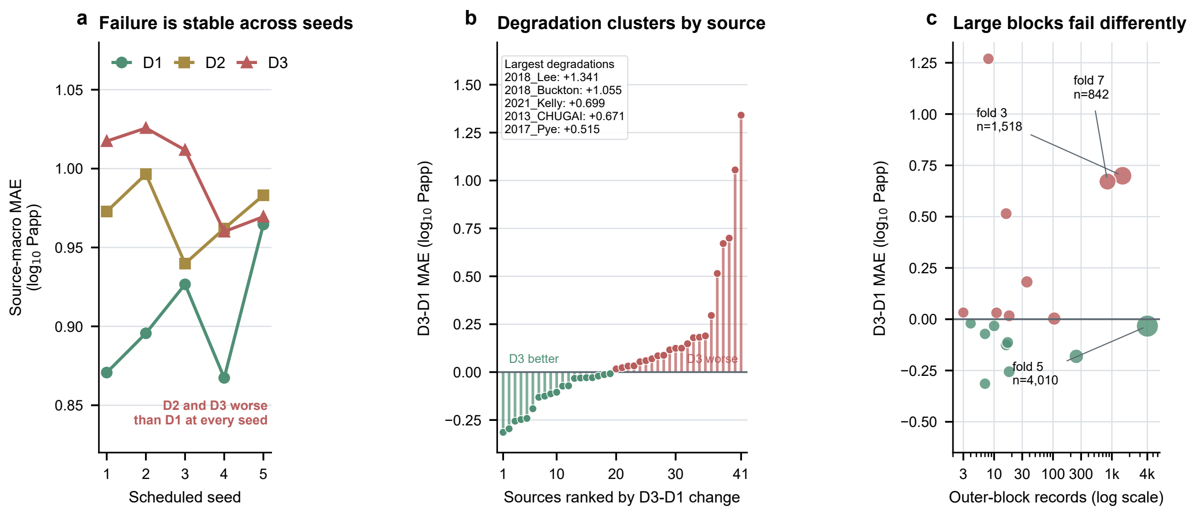

Figure 4: Seed-stable but regime-dependent failure of the fixed D3 candidate. **a,** Source-macro MAE for D1, D2 and D3 across all five scheduled seeds. D2 and D3 are worse than D1 at every corresponding seed. **b,** D3-minus-D1 source-level MAE changes for all 41 sources, ranked from improvement to degradation. Values are five-seed means; green indicates lower error for D3 and red indicates higher error. The inset lists the five largest degradations. **c,** D3-minus-D1 block-level MAE change against the record count of every outer block on a logarithmic axis. Point area also represents record count. Folds 3 and 7 deteriorate strongly, whereas the largest block, fold 5, is slightly improved, demonstrating that failure is not a monotonic consequence of block size. Panels b and c are prespecified descriptive diagnoses and were not used to select a replacement model. Source data are provided in the figure source-data directory.

Discussion

Within this dataset and tested model ladder, estimated cyclic-peptide permeability performance changed more across the two evaluation procedures than across the prespecified representation changes. Holding the curated population, endpoint, baseline architecture, and seeds fixed, joint source/analogue blocking increased source-macro MAE by 0.537 log10(Papp). The lower bootstrap limit was 0.324, and both small-cluster sensitivity intervals remained above zero. The central finding is therefore not that D0 has one intrinsic performance value. Rather, its apparent accuracy depends on whether the evaluation asks for interpolation within familiar sources and analogue neighborhoods or transfer beyond both. Because fold-specific training sizes also changed, the present comparison does not yet apportion the difference between smaller training populations and removal of dependence paths.

This distinction is an estimand choice, not merely a preference for a harder benchmark. Molecule-random cross-validation samples training and test records from the same compiled mixture and can be appropriate when future records are expected to resemble that mixture. It is insufficient when the intended use involves a new assay source, medicinal-chemistry program, or analogue family. In that setting, source-specific measurement offsets and local chemical relationships are part of the deployment shift. A model that benefits from either dependency during evaluation may still be useful for interpolation, but its random-split error should not be interpreted as evidence of out-of-block transfer.

The source-stratified diagnostics sharpen this interpretation. Most sources showed a positive joint-minus-random error gap, while source-level signed bias became substantially broader under joint shift. The calibration slope of 0.595 indicates that the joint-block predictions were compressed toward the training-domain mean. This pattern is consistent with both source offsets and chemical-domain changes affecting transfer. It does not identify their separate causal contributions, because source and analogue paths were intentionally closed together.

The joint-block construction makes the combined dependence paths explicit. Source-only blocking would leave a close series connected through another source, whereas analogue-only blocking would leave shared provenance across chemically separated components. Joining sources and analogue components in a bipartite graph closes both paths by construction. This differs from treating source and chemical similarity as two separate sensitivity analyses. The price of the stronger boundary is a smaller effective sample: 6,895 analytical records became 18 indivisible blocks, of which three contained 92.39% of all records. The imbalance is scientifically inconvenient but informative. It shows that the public compilation contains much less independent transfer evidence than its molecular row count suggests.

The source label is nevertheless a provenance proxy, not a structured assay-condition variable. The downloaded table does not provide harmonized fields for PAMPA pH, membrane composition, incubation time, temperature, or donor/acceptor conditions. A source effect can therefore combine assay implementation, reporting convention, campaign chemistry, and unrecorded protocol differences. Joint blocking protects the evaluation boundary against this combined dependence, but it cannot attribute the observed error gap to any one experimental factor.

The threshold reconstruction supports the geometry of this argument without converting it into a performance sensitivity claim. Relaxing or tightening the Tanimoto threshold changed the molecular component count more than eightfold, yet yielded only 15-20 joint blocks. The three largest blocks still contained more than 92% of all records, and the inverse-Simpson effective block count at the primary threshold was only 2.48. Source-provenance closure therefore preserved the central concentration problem across the prespecified threshold range. Complete nested retraining would be required to compare predictive performance under those alternative partitions.

The model ladder further indicates that complexity did not compensate for this geometry. A descriptive outer-training mean predictor had lower source-macro MAE than D0, D1–D3, and the nested-selected classical procedure. This does not mean molecular structure is irrelevant to permeability. The mean predictor cannot rank analogues or support chemical optimization. Its performance instead suggests that, under the present metric and block structure, errors introduced by source and series transfer can outweigh the gains learned from molecular variation. The divergence between row-micro and source-macro rankings supports the same interpretation: a model can fit record-rich regimes reasonably well while performing poorly after every source receives equal weight.

Balanced loss was the only DMPNN modification with a clear descriptive reduction in source-macro MAE, but its gains were heterogeneous. D1 improved 22 of 41 sources and performed better than D0 in 7 of 18 blocks. Descriptor augmentation then erased large gains in several sources and two large blocks, while the largest block remained comparatively stable. This pattern is compatible with descriptor distributions or representation interactions changing across regimes. It is not evidence that medicinal-chemistry descriptors are intrinsically harmful. The frozen experiment did not isolate scaling, redundancy, or optimization as causal mechanisms, and using the revealed OOF results to redesign D2 would invalidate the benchmark as confirmatory evidence.

The negative D3 result also demonstrates why candidate identity must be fixed before evaluation. After outcome release, D1 had the best point estimate among D0–D3 and was slightly better than the nested-selected classical procedure. Replacing D3 with D1 at that stage would select the hypothesis on the same data used to test it. Reporting the original candidate and all five failed criteria preserves the distinction between a descriptive lead and confirmatory support. A future D1-centered or descriptor-revised model would require a new exploratory phase, a frozen hypothesis, and fresh untouched outcomes before a confirmatory claim.

Under-coverage extends the generalization problem from point prediction to uncertainty. The nominal 90% D3 interval covered only 77.0% of OOF predictions despite a mean full width of 2.614 log10(Papp). Fold-local residuals were generated without outer-test leakage, and checkpoint hashes confirmed that the reporting pass performed inference only. The remaining explanation is statistical rather than procedural: residuals observed among the outer-training blocks did not adequately represent errors in the omitted block. Calibration methods for molecular prediction depend on the relation between calibration and deployment examples [20, 21, 22]. When an entire connected regime is held out, exchangeability is doubtful, so empirical residual intervals should be presented as diagnostics rather than prospective guarantees.

The curation audit adds a separate endpoint limitation. All 372 rows carrying detection-limit information were excluded under the frozen rule. Their database values and notes did not encode one common quantitative bound: 259 were assigned or reported database floors, 40 stated upper limits, 19 had no reportable value, 9 indicated non-detection or a limit of detection, 7 reflected solubility or unperformed testing, and 38 had other notes. Only two excluded rows were linked to a retained source-structure group with an uncensored replicate. Treating the remaining heterogeneous sentinels as ordinary continuous permeability values, or imposing one universal censoring threshold, would therefore create an unsupported numerical assumption rather than a principled sensitivity analysis.

The findings complement earlier efforts to replace random splitting with scaffold, temporal, source-held-out, canonical-group, representation-cluster, or simulated out-of-distribution evaluations [10, 12, 13, 18, 7, 19, 8]. Geylan et al. provide the closest provenance-aware precedent by holding out large sources and separately evaluating source and canonical-group splits; PeptideCLM and MultiCycPermea add representation-cluster and chemical-OOD evaluations. The contribution here is narrower than declaring one split universally correct or claiming the individual ingredients as new. In the primary literature checked through 10 August 2026, we did not identify a cyclic-peptide permeability benchmark that closed complete sources and transitive analogue families as joint connected components throughout nested development (Supplementary Table S6). This is a search-limited distinction. Similarity-aware tools such as DataSAIL formalize the broader principle that the split should control the dependencies relevant to the intended claim [16].

Several practical recommendations follow. First, compiled property datasets should retain source identifiers rather than discarding them after molecular standardization. Second, split reports should state the numbers and sizes of independent groups, not only the numbers of rows and folds. Third, model selection, early stopping, preprocessing, and calibration must inherit the same grouping boundary as the outer test. Fourth, source-macro and row-micro metrics should be reported together when sources differ markedly in size. Fifth, simple mean and classical baselines remain necessary under distribution shift, even when the scientific focus is a deep molecular model. Finally, negative prespecified candidates and failed calibration criteria should remain visible because they constrain what the benchmark supports.

Several limitations bound the conclusions. The analysis used one compiled cyclic-peptide PAMPA resource and one predefined model/comparator ladder. The 18 primary blocks were extremely imbalanced, with an inverse-Simpson effective count of 2.48, so the estimates describe the observed evidence geometry rather than an equal-probability population of future sources. Source provenance was available, but harmonized assay-condition fields were not; source blocking therefore controls a composite dependency rather than a named experimental factor. The H1 comparison also changed the outer training-size profile; a size-profile-matched molecule-random control and source-only/analogue-only boundary ladder were frozen as post-confirmatory analyses and remain pending in this version. The originally planned chirality-aware Murcko arm was not completed and is not silently represented by the transitive analogue analysis. No prospective matched-assay PAMPA panel was available, and the study therefore makes an evaluation claim rather than a prospective discovery claim. Threshold reconstruction showed similarly concentrated geometry at 0.70 and 0.90, but did not provide valid alternative-partition performance estimates. Such estimates would require complete exploratory nested retraining and could not replace the primary confirmatory result. Finally, the failure-structure analysis localizes where D2/D3 deteriorated but does not identify a molecular or optimization mechanism.

The strongest next test would be a prospective, matched-protocol PAMPA panel selected before outcomes are known. Such a panel could determine whether the source/analogue-block estimate better predicts performance on genuinely new chemistry and whether a newly frozen model provides enrichment within a controlled series. It would not revise the present H2 verdict. The current contribution is evidence that the molecule-grouped estimate for the tested DMPNN did not carry over to the fully nested joint source/analogue boundary in this dataset.

Conclusion

This prospectively governed retrospective study tested one DMPNN and a fixed comparator ladder on one public cyclic-peptide PAMPA benchmark under simultaneous exclusion of complete data sources and transitive analogue families. Source-macro MAE was 0.467 under molecule-grouped pseudorandom cross-validation and 1.005 under joint-block evaluation, with a joint-minus-random difference of 0.537 and all reported block-level sensitivity intervals above zero. The fixed weighting-plus-descriptor candidate did not close this gap: D3 failed all five confirmatory criteria, ranked below the nested-selected classical procedure, and achieved only 77.0% slot-pooled coverage for nominal 90% empirical intervals. The H1 contrast does not yet separate dependence removal from fold-specific training-set reduction.

The resulting contribution is an evaluation framework and a bounded finding about this dataset and model ladder, not a new predictive state of the art or a universal causal claim about splitting. Here, the number and structure of independent source/analogue blocks were more consequential to the reported estimate than the tested representation additions. Future cyclic-peptide benchmarks should preserve provenance, close transitive chemical-dependence paths, keep model development within grouped training data, and report simple baselines alongside complex models. Prospective matched-protocol PAMPA remains necessary to establish performance on genuinely new chemistry, but it cannot retrospectively convert the unsupported D3 hypothesis into a positive result.

Data and Software Availability

The study used the CycPeptMPDB PAMPA table associated with the upstream database report (`https://doi.org/10.1021/acs.jcim.2c01573`). The exact raw file used locally contained 7,298 rows and had SHA-256 `02da1cfc18a92b3ae6e70152445b23c05ce6bb0b6ed10fc7c9e141fbd9462fde`. Because permission to redistribute the upstream structures and permeability labels has not yet been established, the release candidate excludes that table and provides its expected filename, hash, schema, acquisition route, and deterministic reconstruction instructions. Release-safe row and curated-group manifests expose identifiers, inclusion/exclusion accounting, source labels and fold membership without structures or endpoint values. The package also contains the pinned baseline provenance, protocol and deviation timeline, exact split manifests, machine-readable analysis source data, integrity summaries, reporting code, and file-level checksums. It is available at **`https://github.com/schwarzeteeguo-creator/scaffoldseal-cp-evaluation`** and will be archived at **[Zenodo DOI pending]** before submission. This draft must not be submitted until the author list, repository license, stable data-access route, and both public identifiers have been verified against the exact versioned archive.

Supporting Information

Supporting Information: detailed curation, modelling, validation and reproducibility methods; Supplementary Figures S1–S3; Supplementary Tables S1–S11; and the protocol, release and source-data inventory (PDF).

Author Contributions

[CRediT roles and author initials to be supplied.]

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Conflict of Interest

The authors declare **[no competing financial interest / statement to be supplied]**.

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